

Technical Datasheet Input Module with Modbus TCP over Ethernet Interface

This IO module communicates via Ethernet Port. The port can drive distances up to max 80 meters without the use of any repeater (*this is as per the ethernet LAN guidelines*). The Modbus TCP Digital IO module is sturdy, low power usage and easy to use.

16 Port DI Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. the board also has '**reverse polarity**' protection both for input and power connector.

the communication port also has signal isolation and additional jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw Terminal Block and maximum 2.5 sq mm cable connector
Dimensions	110 mm B x 110 mm L x 50 mm H
Power	Input Power – 12 VDC or 24 V AC/DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing



Certifications

DI Inputs –

Channels	16
Sense Voltage	4 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Configuration Settings: -

Communication Speed	10/100 Base-T Ethernet Compatible Interface
Data Bits	8
Parity	None
CRC	1
Stop bit	Yes
Slave ID	Configurable through Web portal
Function code DI	0x02 Read discrete Input
DI Register Address	0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15.

Configuration Guide – Ethernet Modbus TCP/IP IO Module:-

1. Open a web browser (Google Chrome, Microsoft Edge, Firefox, etc.).
2. In the browser's address bar, type the default IP address:
192.168.0.160
3. Press **Enter**. A message will appear saying:
"Please go to <http://192.168.0.160/setup>."
4. Copy this link (<http://192.168.0.160/setup>) and paste it into a new browser tab.
5. Press **Enter** again to open the **Setup / Configuration Page** of the device.

To Reset the Device to Default Settings

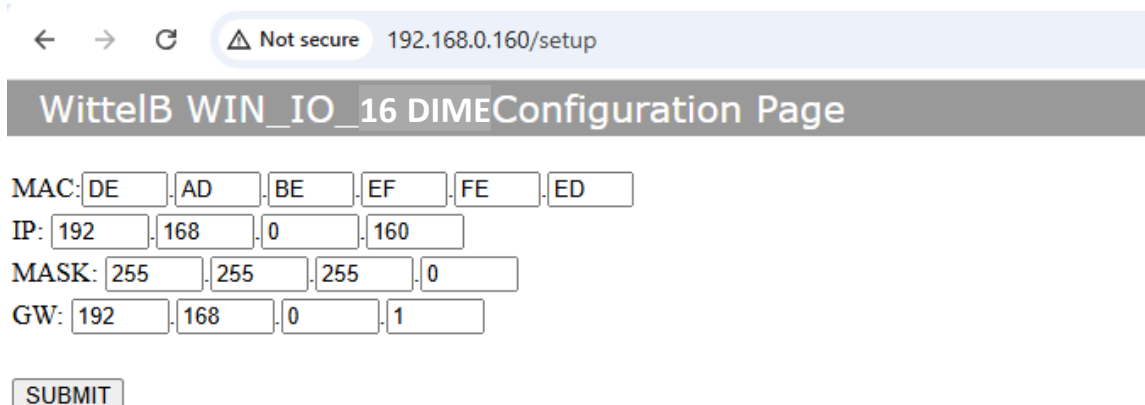
- Place the jumper on the designated **Reset position** on the device.
- Press the **Reset button** once.
- The module will reboot and revert to its default IP address: **192.168.0.160**.

To Change the IP Address

- On the **Setup / Configuration Page**, enter the new IP address you wish to assign.
- Click **Submit** to save the new settings.
- The device will automatically apply the new IP address.

Note:

- Ensure your PC's IP address is in the **same network range** (for example, **192.168.0.xxx**) when accessing the module.
- After changing the IP, you must use the **new address** to log in again.

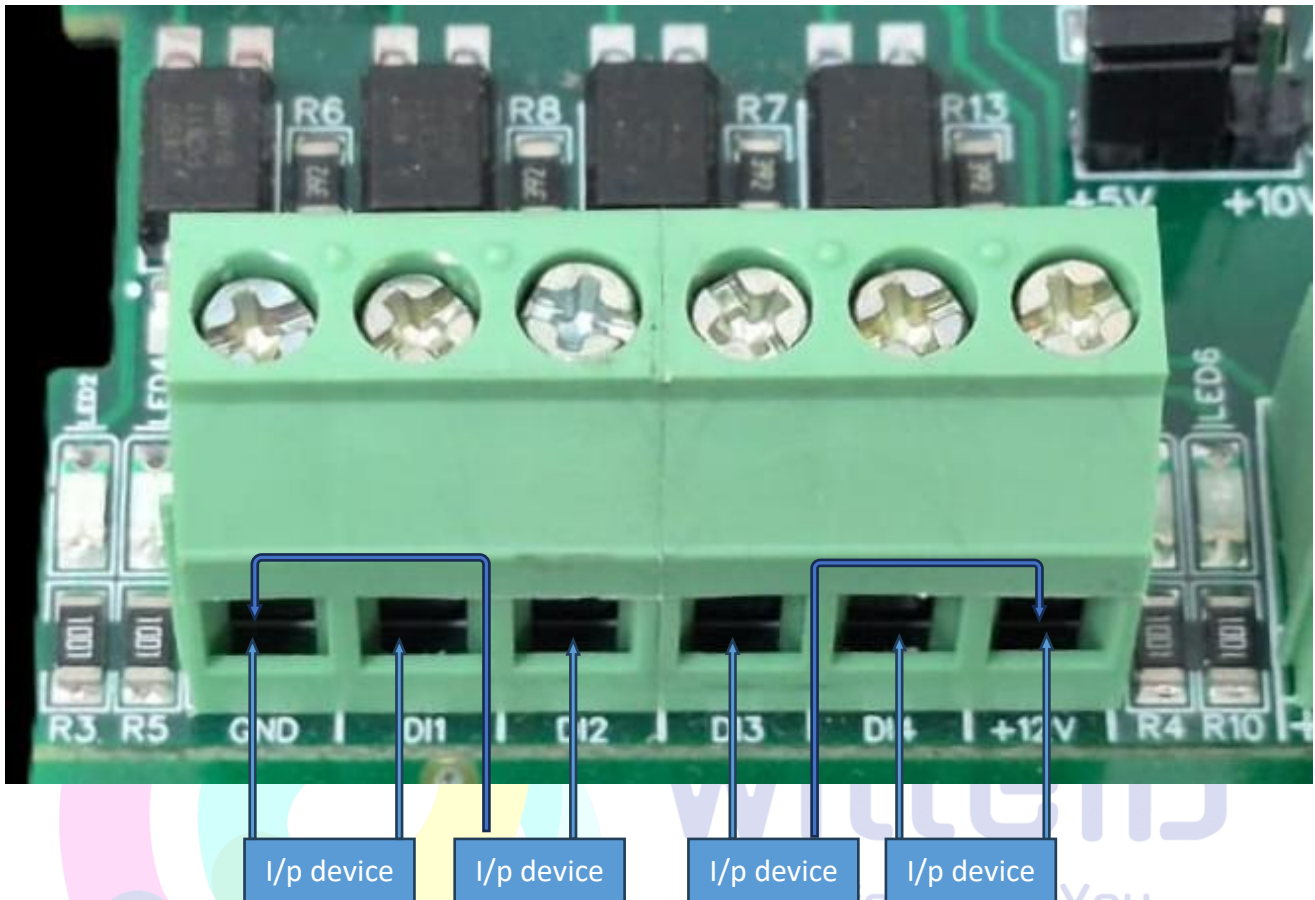


The screenshot shows a web browser window with the address bar displaying "192.168.0.160/setup". The page title is "WittelB WIN_IO_16 DIME Configuration Page". The form contains the following fields:

MAC:	DE	AD	BE	EF	FE	ED
IP:	192	168	0	160		
MASK:	255	255	255	0		
GW:	192	168	0	1		

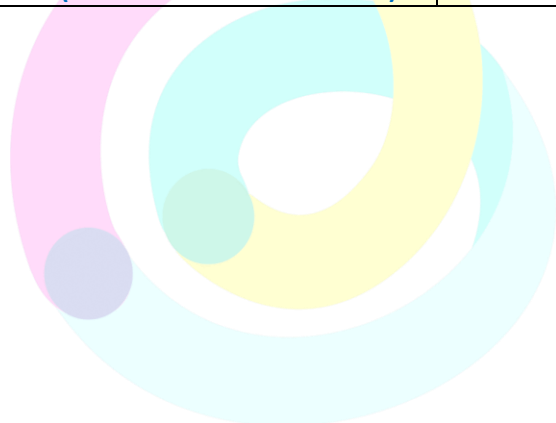
Below the fields is a **SUBMIT** button.

DI



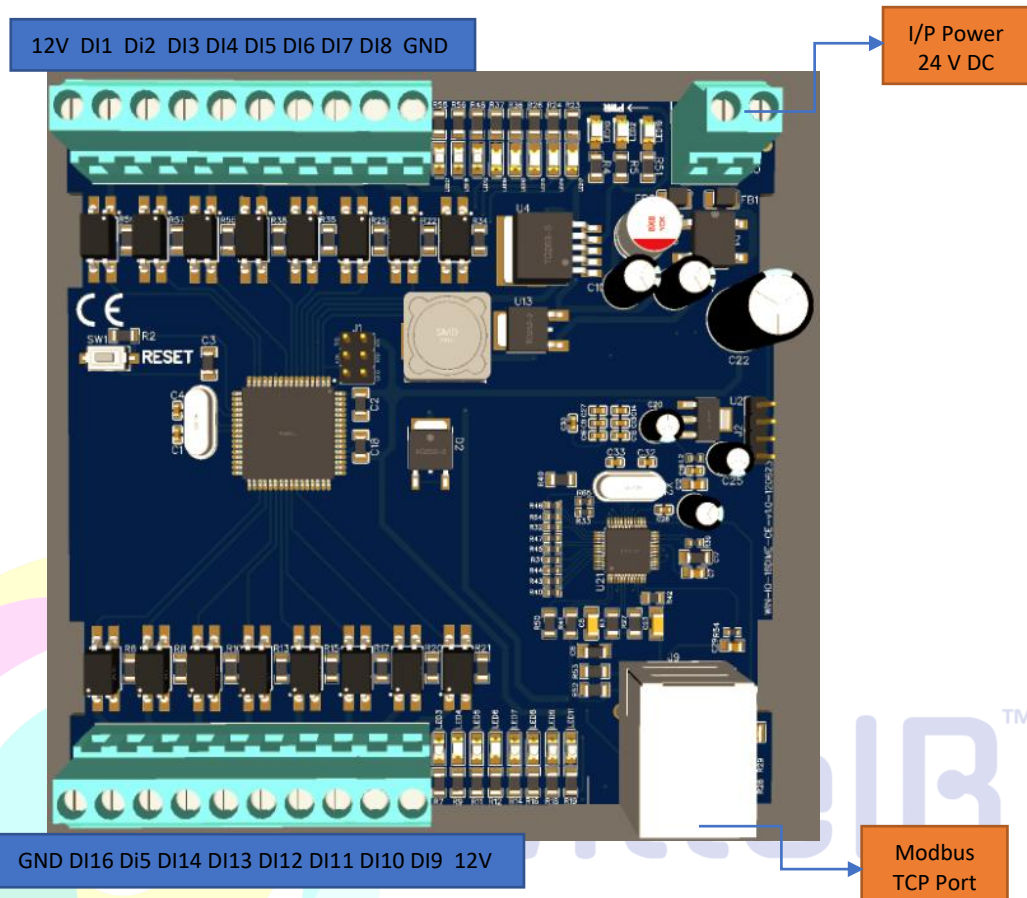
- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DI 1(With & Without Potential)	10001	0X01	TCP/IP	1 Bit Boolean
2	DI 2(With & Without Potential)	10002	0X01	TCP/IP	1 Bit Boolean
3	DI 3(With & Without Potential)	10003	0X01	TCP/IP	1 Bit Boolean
4	DI 4(With & Without Potential)	10004	0X01	TCP/IP	1 Bit Boolean
5	DI 5(With & Without Potential)	10005	0x01	TCP/IP	1 Bit Boolean
6	DI 6(With & Without Potential)	10006	0X01	TCP/IP	1 Bit Boolean
7	DI 7(With & Without Potential)	10007	0X01	TCP/IP	1 Bit Boolean
8	DI 8(With & Without Potential)	10008	0X01	TCP/IP	1 Bit Boolean
9	DI 9(With & Without Potential)	10009	0X01	TCP/IP	1 Bit Boolean
10	DI 10(With & Without Potential)	10010	0X01	TCP/IP	1 Bit Boolean
11	DI 11(With & Without Potential)	10011	0X01	TCP/IP	1 Bit Boolean
12	DI 12(With & Without Potential)	10012	0X01	TCP/IP	1 Bit Boolean
13	DI 13(With & Without Potential)	10013	0X01	TCP/IP	1 Bit Boolean
14	DI 14(With & Without Potential)	10014	0X01	TCP/IP	1 Bit Boolean
15	DI 7(With & Without Potential)	10015	0X01	TCP/IP	1 Bit Boolean
16	DI 8(With & Without Potential)	10016	0X01	TCP/IP	1 Bit Boolean



wittelBTM
 Our focus is You

Product Image:



For more details: <http://www.youtube.com/@wittelb>

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